Round 2 Technical Assessment

AI Innovation Hackathon 2026 – Build Real-World AI Solutions

Challenge Title

AI Teacher: Build a Human-Like AI Educator That Teaches Through Video

Problem Statement

Traditional digital learning platforms generally provide pre-recorded lectures or text-based AI assistants. However, these systems often fail to provide the personalized interaction and adaptive teaching approach of a real teacher.

Your challenge is to design and develop an **AI-powered virtual teacher** capable of understanding learning material, creating personalized lessons, and teaching students through an AI-generated video experience.

The AI Teacher should be able to take an uploaded **book, textbook, PDF, notes, presentation, research paper, or other educational material**, or accept a topic directly from the student, and transform it into a structured and personalized teaching session.

The system should not function as a basic question-answer chatbot. It should demonstrate the behavior of a real teacher by understanding the learner, planning the lesson, explaining concepts, providing examples, asking questions, evaluating responses, identifying areas of difficulty, and adapting the teaching approach.

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1. Objective

Build a working AI Teacher that can:

- Understand uploaded educational content.
- Teach any user-provided topic.
- Generate a structured lesson plan.
- Explain concepts according to the learner's level.
- Adapt the lesson according to the available learning time.
- Teach in multiple languages.
- Present lessons through a human-like AI avatar and voice.
- Generate appropriate visual explanations.
- Interact with the student during the lesson.
- Evaluate the student's understanding.
- Identify misconceptions and provide alternative explanations.
- Track learning progress.
- Recommend what the student should learn or revise next.

2. User Scenario

A student uploads a textbook or learning material and provides an instruction such as:

"I am a beginner. Teach me Chapter 4 in 20 minutes. Explain it in Hindi using simple examples. Ask me questions during the lesson and test me at the end."

The AI Teacher should process the material and determine:

1. What needs to be taught.
2. Which concepts should be covered first.
3. How deeply each concept should be explained.
4. Which examples or visuals should be used.
5. When the student should be questioned.
6. Whether the student has understood the concept.
7. Whether the lesson needs to be simplified or expanded.
8. What should be taught next.

The final experience should resemble an actual personalized teaching session rather than a conventional chatbot interaction.

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3. Learning Material Processing

The system should support learning from user-provided materials such as:

- Books
- Textbooks
- PDF documents
- Lecture notes
- DOC/DOCX files
- PPT/PPTX files
- Research papers
- Course material
- Other text-based educational resources

The AI should be capable of identifying relevant chapters, sections, concepts, definitions, examples, and other useful information from the uploaded material.

Participants are encouraged to implement **Retrieval-Augmented Generation (RAG)** or another suitable knowledge-grounding approach.

The system should minimize unsupported or hallucinated information when answering questions related to uploaded material.

4. Topic-Based Learning

The AI Teacher must also be capable of teaching topics without uploaded material.

For example:

"Teach me Artificial Intelligence from the beginning."

"Explain Newton's Laws to a Class 8 student."

"Teach me React for a technical interview."

The system should generate an appropriate learning structure based on the requested topic and learner profile.

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5. Human-Like Teaching

This is the primary requirement of the assessment.

The AI Teacher should follow a teaching process similar to a human educator:

Understand → Plan → Explain → Demonstrate → Question → Evaluate → Adapt → Continue

The teacher should:

- Introduce the topic.
- Explain concepts progressively.
- Use appropriate examples.
- Ask questions at suitable points.
- Analyze the student's responses.
- Correct mistakes constructively.
- Re-explain difficult concepts.
- Adjust difficulty.
- Confirm understanding before moving forward.

A system that only generates answers to questions will not be considered a complete solution.

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6. Personalized Teaching

The AI Teacher should adapt its teaching according to the learner.

The learner may specify:

- Educational level.
- Existing knowledge.
- Learning objective.
- Preferred teaching style.
- Preferred language.
- Available time.
- Desired depth.

For example:

Beginner:

Simple terminology, analogies and fundamental concepts.

Intermediate:

More technical explanations and practical examples.

Advanced:

Detailed concepts, technical terminology, mathematics, implementation details and advanced examples where appropriate.

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7. Time-Based Learning

The student should be able to specify the amount of time available.

Examples:

5 minutes:
Provide a concise explanation of the most important concepts.

20 minutes:
Create a structured lesson covering the key concepts with examples.

60 minutes:
Provide a deeper lesson with explanations, examples, questions and assessment.

7 days:
Generate a personalized learning and revision plan.

The AI should change the lesson structure and depth according to the specified timeframe.

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8. Multilingual Teaching

The AI Teacher should support multiple languages.

Students should be able to select their preferred teaching language or request a language naturally during the conversation.

For example:

"Explain this topic in Hindi."

"Now explain it in English."

"Mujhe ye Hinglish mein simple example ke saath samjhao."

The system should maintain the context of the lesson when the language changes.

The uploaded material may also be in a different language from the student's preferred teaching language.

For example:

English textbook → Hindi teaching

or

Hindi textbook → English teaching

Participants supporting multiple Indian and international languages will receive additional consideration.

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9. AI Teaching Video

The AI Teacher must present the lesson through a video-based teaching experience.

The generated teaching video should ideally include:

- Human-like AI avatar.
- Natural voice.
- Spoken explanation.
- On-screen text.
- Diagrams and illustrations.
- Relevant images.
- Subject-specific visualizations.
- Mathematical or technical demonstrations where applicable.

The objective is to create the experience of attending a lesson conducted by an AI teacher.

Simply placing a talking avatar in front of generated text will not be considered sufficient for a strong implementation.

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10. Subject-Aware Visual Explanation

The AI should use appropriate visuals according to the subject.

For example:

Mathematics

- Equations
- Graphs
- Step-by-step solutions

Physics

- Diagrams
- Formulas
- Processes
- Simulations or visual demonstrations

Biology

- Labeled diagrams
- Biological processes
- Structures

History

- Timelines
- Maps
- Events

Programming

- Code
- Output
- Execution flow
- Architecture diagrams

Participants should demonstrate how their system determines which visual representation is appropriate for the topic.

11. Interactive Learning

The AI Teacher should not continuously deliver a monologue.

It should periodically interact with the student.

Possible interactions include:

- Conceptual questions.
- Multiple-choice questions.
- Short-answer questions.
- Problem-solving questions.
- Application-based questions.
- "Explain in your own words" questions.

The student's responses should influence the subsequent teaching.

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12. Misconception Detection and Adaptive Teaching

The system should identify when a student does not understand a concept or provides an incorrect answer.

For example:

Teacher:
"What happens to current if resistance increases while voltage remains constant?"

Student:
"Current increases."

The AI should identify the misunderstanding and provide an appropriate explanation rather than simply marking the answer as incorrect.

An advanced system may:

1. Identify the misconception.
2. Explain the underlying concept again.
3. Use a different analogy.
4. Provide another example.
5. Ask a new question.
6. Re-evaluate the student's understanding.

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13. Assessment and Feedback

After completing a lesson, the AI Teacher should evaluate the student's understanding.

The system may generate:

- Quiz questions.
- Conceptual questions.
- Practical problems.
- MCQs.
- Short-answer questions.

The AI should provide a learning report containing, where applicable:

- Score.
- Concepts understood.
- Weak areas.
- Incorrect concepts.
- Recommended revision.
- Suggested next topic.

Example:

Topic: Electricity

Score: 80%

Strong Areas: Current, Voltage

Needs Improvement: Resistance, Ohm's Law

Recommendation: Revise Ohm's Law and complete two additional practice problems.

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14. Student Learning Profile

Participants are encouraged to maintain a learner profile containing information such as:

- Topics studied.
- Progress.
- Assessment scores.
- Weak concepts.
- Strong concepts.
- Learning history.
- Current learning path.

The AI should use this information to personalize future teaching sessions.

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15. AI-Generated Learning Path

For broad topics, the AI should be able to create a structured learning path.

For example:

Machine Learning

1. Python Fundamentals
2. Mathematics for ML
3. Data Processing
4. Supervised Learning
5. Unsupervised Learning
6. Model Evaluation
7. Neural Networks
8. Advanced Machine Learning

The AI should be able to guide the student through the learning path based on their progress.

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16. Technology

Participants may use appropriate:

- Large Language Models
- Generative AI
- Machine Learning models
- RAG systems
- Vector databases
- Speech-to-Text
- Text-to-Speech
- AI Avatar technologies
- Computer Vision
- Generative media technologies
- Web/mobile technologies
- Cloud services
- Open-source models and frameworks

Teams must clearly disclose significant third-party APIs, models, libraries, and services used in their solution.

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17. Mandatory Requirements

A valid submission should demonstrate at least:

1. Learning from uploaded material or documents.
2. Topic-based teaching.
3. AI-generated lesson structure.
4. Personalized teaching.
5. Human-like teaching interaction.
6. Video-based AI Teacher presentation.
7. AI voice.
8. Human-like AI avatar.
9. Multilingual capability.
10. Student questioning and assessment.
11. Adaptive response to student performance.
12. Working application/prototype.

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18. Advanced Features

The following are not mandatory but can strengthen the submission:

- Real-time conversational teaching.
- Multiple teacher personalities.
- Emotion-aware interaction.
- Long-term student memory.
- Automatic study planner.
- Exam preparation mode.
- Revision mode.
- Flashcard generation.
- Automatic notes.
- Concept maps.
- Coding demonstration.
- Interactive diagrams.
- Personalized homework.
- Learning analytics.
- Offline/local AI models.
- Accessibility features.
- Multiple AI teacher characters.

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19. Evaluation Criteria

Evaluation Area	Weightage
Human-Like Teaching and Adaptation	20

AI/ML and LLM Implementation	15
RAG and Knowledge Grounding	15
AI Teaching Video Generation	15
Multilingual Capability	10
Voice and AI Avatar	10
Innovation and Originality	5
User Experience and Interface	5
Documentation and Technical Presentation	5
Total	**100**

Evaluation Focus

The jury will primarily evaluate whether the solution demonstrates **genuine AI-driven teaching capability**.

A basic chatbot, a static video, or a talking avatar reading a generated script will not be considered equivalent to an adaptive AI Teacher.

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20. Submission Requirements

Each participant/team must submit:

Source Code

GitHub repository or ZIP containing the project.

Project Documentation

The documentation should include:

- Problem statement.
- Solution overview.
- Key features.
- System architecture.
- AI/ML models used.
- RAG implementation.
- Prompt/agent architecture.
- Personalization approach.
- Assessment methodology.
- Multilingual implementation.
- Voice implementation.
- Avatar/video generation approach.
- APIs and third-party services.
- Setup instructions.
- Deployment instructions.
- Known limitations.

Working Prototype

A functional application or deployed demonstration.

Demo Video

Recommended duration: **3–7 minutes**.

The demo should show the complete journey:

Upload/Topic → Lesson Planning → AI Teaching Video → Student Interaction → Adaptation → Assessment → Learning Feedback

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Final Challenge

Build the AI Teacher of the Future

Participants must complete the following two tasks:

Task 1 — AI Teaching Video

Build an AI Teacher that can take **any topic or uploaded learning material** such as a book, textbook, PDF, notes, or other educational content and create a **personalized teaching video**.

The AI Teacher should:

- Understand the provided material or topic.
- Create a structured lesson.
- Adapt the explanation according to the learner's level and available time.
- Explain concepts using a human-like AI avatar and natural voice.
- Use relevant visual explanations such as diagrams, examples, formulas, images, or code where appropriate.
- Support multiple languages.
- Deliver the lesson as an engaging educational video rather than a simple text response.

Task 2 — Interactive & Adaptive AI Teacher

Extend the AI Teacher so that it can **interact with the learner and adapt the lesson based on their responses**.

The AI Teacher should:

- Ask questions during the lesson.
- Understand and evaluate student responses.
- Identify incorrect answers and knowledge gaps.

- Re-explain concepts when the student is struggling.
- Change the difficulty level based on student performance.
- Answer follow-up questions while maintaining lesson context.
- Conduct a final assessment or quiz.
- Provide personalized feedback and recommend what the student should learn or revise next.

Do not build a chatbot that simply answers questions.

Build an AI Teacher that understands, explains, interacts, adapts, and actually teaches.

Short Demo Ai Teacher video: https://drive.google.com/file/d/1y0yXus5J9iS9m-fg5QI1qjqvgA55U_Kw/view?usp=drive_link

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